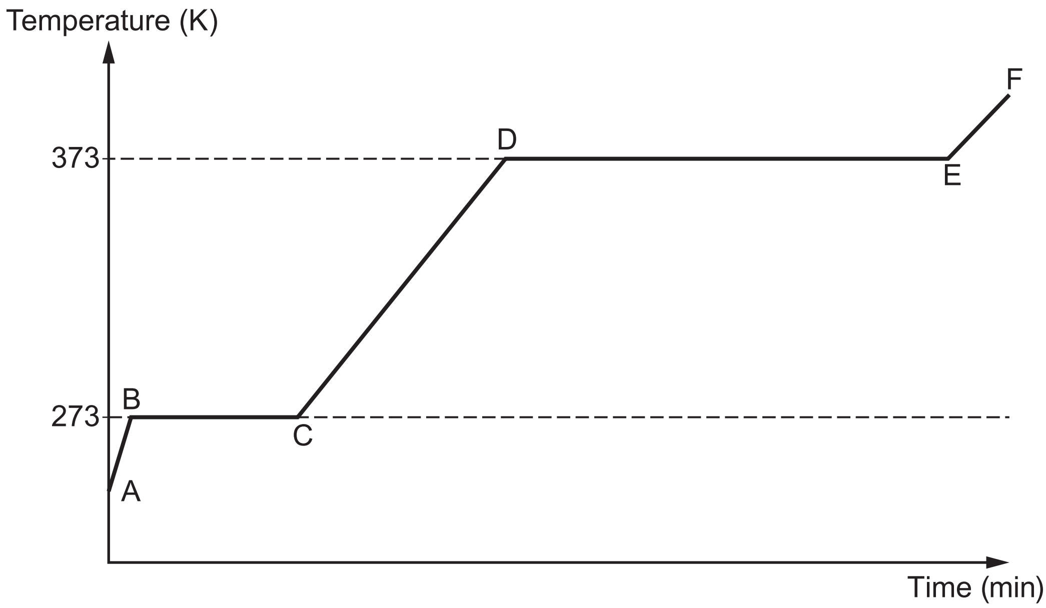
6. The graph below shows how the temperature of a block of ice changes over time as it is heated.



(a) Refer to the graph to explain the changes in temperature and changes in state of the block of ice in terms of the behaviour of molecules. [6 QER]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

Examiner only

(b) The specific latent heat, L, of fusion of ice is 336 000 J/kg.

(i) Explain what is meant by this statement. [2]

....

....

....

(ii) The ice block has a mass of 800 g. Its initial temperature is 250 K.
Use equations from page 2 to calculate the energy required to raise its temperature to 273 K and completely melt the ice block at 273 K.
The specific heat capacity, c, of ice is 2 030 J/kg K. [5]

Energy = ... J

13